Product Requirement Document

AI-Powered Family Location Tracker

1. Problem Statement & Target Users

Problem Statement

Families want to ensure the safety and coordinate the daily movements of their members (school commutes, office travel, elderly care). However, existing solutions lock essential features behind expensive monthly subscriptions, drain battery life, and raise privacy concerns by selling user data. Furthermore, manually scrolling through past location history on a map is tedious.

Target Users

- **Parents:** Need to ensure children safely reach school, college, or extra-curricular activities without intrusive micromanagement.
- **Adult Caregivers:** Monitoring the safety, routines, and unexpected deviations of elderly parents.
- **Coordinating Families:** Family members who want a quick, conversational way to check schedules and locations without manual map scrubbing.

2. Core Features & Functionality

- **Live Location Tracking:** Real-time GPS tracking displayed on an interactive, lightweight OpenStreetMap interface.
- **Historical Data Logging:** Automatic background logging of location coordinates saved to a cloud database (MongoDB) with a 30-day rolling retention to optimize storage.
- **AI Movement Assistant:** A natural language chat interface using Google AI Studio where users can ask questions about location history. The AI parses database coordinates and translates them into readable human summaries.
- **Zero-Cost Architecture:** Strict adherence to 100% free, open-source, and free-tier infrastructure.

3. User Stories & Workflows

Persona	Goal / Need	Expected Outcome
Parent	Open app to check child's live commute.	Instantly see a moving marker of child's location to confirm safe arrival on the bus route.
Spouse	Ask "When did John leave work today?"	Conversational AI response without manually scrubbing through a visual map timeline.
Tracked Member	Background tracking without manual app open.	Automatic, silent GPS logging in background consuming minimal phone battery.

4. Requirements Overview

Functional Requirements

- **GPS Fetching:** Client mobile/web app must fetch background GPS coordinates at optimized intervals.
- **Real-time Broadcast:** Backend must broadcast live coordinate updates to active connected clients via WebSockets.
- **GenAI Integration:** Backend must integrate the Google Gen AI SDK to process user natural language prompts against coordinate histories.
- **Data Lifecycle:** System must automatically purge location records older than 30 days via database TTL indexes.

Non-Functional Requirements

- **Infrastructure Cost:** Strictly **$0.00** recurring operational cost utilizing free-tier services.
- **Latency:** Live location updates must reflect on connected maps within **< 2 seconds**.
- **AI Response Speed:** Natural language queries must return synthesized answers in **< 4 seconds**.
- **Battery Efficiency:** Background location tracking must consume **< 10% battery per day**.
- **Data Isolation:** Location and interaction data must be strictly isolated to the authorized family unit.

5. Success Criteria & Expected Outcomes

- **Continuous Tracking:** Successfully track **4+ family members** continuously without exceeding cloud provider free tiers or triggering billing alerts.
- **AI Query Accuracy:** AI Assistant accurately interprets and answers temporal/spatial queries with **90%+ accuracy**.
- **User Experience:** App responsiveness matches paid alternatives while maintaining zero phone overheating and negligible battery impact.